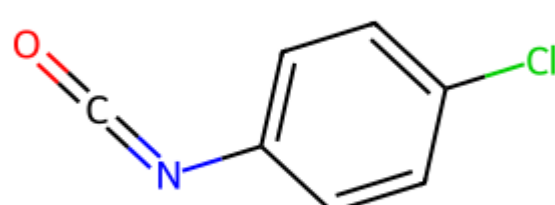


{'analysis\_time': 0.0,  
'first\_solution\_time': 0.174534,  
'num\_calls\_rxn\_model': 0,  
'num\_calls\_value\_function': 0,  
'reaction\_number\_estimate': 0.0,  
'retro\_star\_min\_cost': 2.419118909249997,  
'retro\_star\_mol\_cost': inf,  
'retro\_star\_reaction\_number': 2.419118909249997,  
'retro\_star\_value': 2.419118909249997}

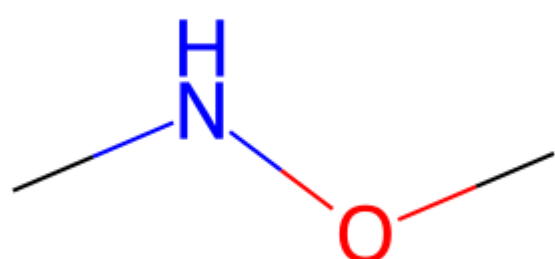


Reaction

{'analysis\_time': 0.174534,  
'first\_solution\_time': 0.266138,  
'num\_calls\_rxn\_model': 1,  
'num\_calls\_value\_function': 0,  
'retro\_star\_min\_cost': 4.804231066786343,  
'retro\_star\_reaction\_number': 4.804231066786343,  
'retro\_star\_rxn\_cost': 0.29437106060257756,  
'retro\_star\_value': 4.804231066786343}



{'analysis\_time': 0.174534,  
'first\_solution\_time': 0.174534,  
'num\_calls\_rxn\_model': 1,  
'num\_calls\_value\_function': 0,  
'retro\_star\_min\_cost': 0.0,  
'retro\_star\_mol\_cost': 0.0,  
'retro\_star\_reaction\_number': 0.0,  
'retro\_star\_value': 4.804231066786343}

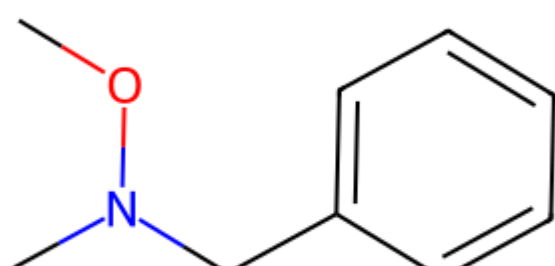


{'analysis\_time': 0.174534,  
'first\_solution\_time': 0.266138,  
'num\_calls\_rxn\_model': 1,  
'num\_calls\_value\_function': 0,  
'reaction\_number\_estimate': 0.0,  
'retro\_star\_min\_cost': 4.509860006183766,  
'retro\_star\_mol\_cost': inf,  
'retro\_star\_reaction\_number': 4.509860006183766,  
'retro\_star\_value': 4.804231066786343}



Reaction

{'analysis\_time': 0.266138,  
'first\_solution\_time': 0.475901,  
'num\_calls\_rxn\_model': 2,  
'num\_calls\_value\_function': 41,  
'retro\_star\_min\_cost': 9.361163261710766,  
'retro\_star\_reaction\_number': 9.361163261710766,  
'retro\_star\_rxn\_cost': 2.4534079827286295,  
'retro\_star\_value': 9.655534322313343}

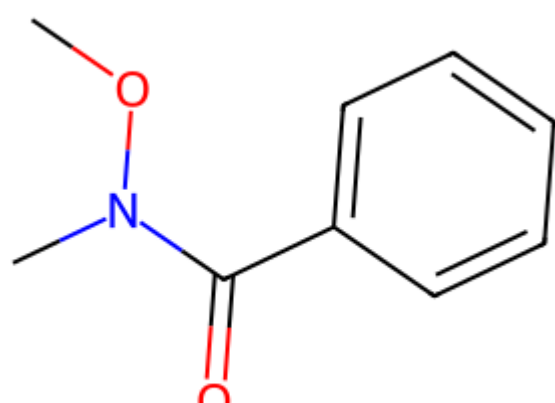


{'analysis\_time': 0.266138,  
'first\_solution\_time': 0.475901,  
'num\_calls\_rxn\_model': 2,  
'num\_calls\_value\_function': 41,  
'reaction\_number\_estimate': 0.0,  
'retro\_star\_min\_cost': 6.907755278982137,  
'retro\_star\_mol\_cost': inf,  
'retro\_star\_reaction\_number': 6.907755278982137,  
'retro\_star\_value': 9.655534322313343}



Reaction

{'analysis\_time': 0.475901,  
'first\_solution\_time': 0.475901,  
'num\_calls\_rxn\_model': 4,  
'num\_calls\_value\_function': 70,  
'retro\_star\_min\_cost': 6.907755278982137,  
'retro\_star\_reaction\_number': 6.907755278982137,  
'retro\_star\_rxn\_cost': 6.907755278982137,  
'retro\_star\_value': 9.655534322313343}



{'analysis\_time': 0.475901,  
'first\_solution\_time': 0.475901,  
'num\_calls\_rxn\_model': 4,  
'num\_calls\_value\_function': 70,  
'retro\_star\_min\_cost': 0.0,  
'retro\_star\_mol\_cost': 0.0,  
'retro\_star\_reaction\_number': 0.0,  
'retro\_star\_value': 9.655534322313343}